

Egor Marin, Forward-Deployed ML Engineer in Cofolding

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SUMMARY

- ML Scientist with a PhD in Structural Biology and 8+ years of research experience
- built and managed ML/DL codebase with data and model registry in a team of scientists and engineers
- specialized in building generative and predictive models for antibody drug discovery and structural biology
- managed complete research data lifecycles, from cryo-EM/crystallography data processing (20TB+) to deploying ML-driven analysis tools
- Core Developer for MDAnalysis, contributing major features like a parallel processing backend (dask/multiprocessing)

EXPERIENCE

Forward-Deployed ML Engineer

Apheris

August 2026 — Present

Berlin, Germany (remote)

Machine Learning Scientist

ENPICOM B.V.

May 2024 — June 2026

Den Bosch, Netherlands

- develop and deploy generative and predictive models for antibody drug discovery using pytorch/lightning and custom discrete diffusion pipelines
- own the end-to-end ML lifecycle, from sourcing and cleaning biochemical data to building reproducible training pipelines with MLflow for experiment tracking
- manage codebase with data and training pipelines, ensuring regular code reviews and code reusability while maintaining modularity and compatibility of the models

Open-source software engineer

MDAnalysis via Google Summer of Code

May 2023 — September 2023

Remote

- designed and wrote backward-compatible parallelization with dask or multiprocessing for the core analysis library

PhD Researcher

University of Groningen

June 2021 — Dec 2023

Groningen, Netherlands

- led a research project as a corresponding author, applying regression-based active learning to accelerate ultra-large library docking, resulting in a publication in J. Chem. Inf. Model
- designed, deployed, and managed a high-throughput data processing infrastructure (CryoSPARC) for 15+ users, processing over 80 cryo-EM projects and 20TB of data
- developed a custom web server (ntfy-cryosparc) to provide personalized notifications, improving workflow efficiency for the research group
- solved 38+ protein structures using cryo-EM and X-ray crystallography, leading to publications in journals including Nature Communications and Science Advances

Junior Research Associate

Moscow Institute of Physics and Technology

March 2017 — September 2021

Moscow, Russia

- managed the end-to-end protein crystallography pipeline for multiple research groups, from sample preparation to synchrotron and XFEL data collection
- processed, refined, and analyzed crystallographic data, successfully determining and depositing 38 protein structures to the PDB
- contributed structural data and analysis that formed the basis for 16 peer-reviewed publications

SKILLS

Technology: python | rust | polars | duckdb | pytorch | jax | numpy | sklearn | pydantic | mlflow | docker | AWS | k8s

Domain: cryoEM | X-ray crystallography | biophysics | drug discovery | cheminformatics | antibodies

SELECTED PUBLICATIONS

For full list, see [google scholar](#).

- Science Advances** July 2025
CryoRhodopsins: a comprehensive characterization of a new clade of microbial rhodopsins from cold environments
- Pre-print on biorxiv** 29 April 2025
Structural basis for no retinal binding in flotillin-associated rhodopsins
- Nature Communications** 4 October 2024
A subgroup of light-driven sodium pumps with an additional Schiff base counterion
- Journal of Chemical Information and Modeling** December 2023
Regression-Based Active Learning for Accessible Acceleration of Ultra-Large Library Docking
- Nature Communications Chemistry** 2 May 2023
Mirror proteorhodopsins
- Nature Communications** August 2022
Structural basis for receptor selectivity and inverse agonism in S1P5 receptors
- Biomolecules** December 2020
Structure-Based Virtual Screening of Ultra-Large Library Yields Potent Antagonists for a Lipid GPCR
- Structural basis of ligand selectivity and disease mutations in cysteinyl leukotriene receptors** December 2019
Nature Communications
- Science Advances** October 2019
Structure-based mechanism of cysteinyl leukotriene receptor inhibition by antiasthmatic drugs

EDUCATION

- University of Groningen** 2019 — 2023
PhD Groningen, Netherlands
- thesis ‘On the methods of studying protein-ligand interaction dynamics’
 - methods: cryoEM, X-ray crystallography, molecular dynamics, protein biochemistry, protein biophysics
 - publications in JACS, Crystal Growth & Design, Journal of Cheminformatics & Modelling
- Moscow Institute of Physics and Technology** 2017 — 2019
MSc in applied mathematics and physics Moscow, Russia
- graduated *summa cum laude*
- Moscow Institute of Physics and Technology** 2013 — 2017
BSc in applied mathematics and physics Moscow, Russia
- graduated *magna cum laude*
 - related coursework: Calculus I-IV, Linear Algebra I-II, Complex Analysis, Differential Equations I-II, Analytical Mechanics I-II, Thermodynamics.

OPEN SOURCE CONTRIBUTIONS & PROJECTS

- Open-source contributions** Jan 2021 — Present
- [polars](#): contributed to polars (issue [#25383](#): extending `replace_many` with `leftmost` option.
 - [polars-distance](#): minor contribution to polars plugin for distance calculation
 - [reciprocalspaceship](#): wrote parser for serial crystallography data into binary dataframe-like class
- MDAnalysis Core Developer** February 2025 — Present
- [MDAnalysis Core Developer](#)
 - wrote a [parallel backend](#) for all analysis classes (dask/multiprocessing)
 - added a [DSSP module](#) for native secondary structure assignment
 - currently working on fast unified [MMCIF parser](#) based on [gemmi](#)
- Self-hosting** Jan 2021 — Present
- self-hosting multiple applications (paperless, llama.cpp, openwebui) for family use under a Tailscale VPN